

**Global Facilities - Design & Construction Standard -
General - Critical Specification Alignment for
Greenfield Fabs and Technology Node Transfer**

ECN Facility: GLOBAL FACILITIES

ECN Area: Standard

Owner(s) ENGINEERING

Target Audience: All Facilities

Document Type: Standard

Document ID (pdf):

Version: 4

A3YRXSD74VDV-57553043-402

ECN: 101161272

Last Modified: 03/11/24

[ECN History](#)

[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document lists the Global Critical Specifications for the Facilities Process systems at Front-End Manufacturing Sites, as determined by the Global Spec Alignment project.
- 1.2 These Global Front-End Spec Alignment Critical Specifications have been developed and approved by Global Manufacturing teams TD and OCT, with additional review and approval by GFAC teams.
- 1.3 The objectives of the Global Critical Specifications are as follows:
 - 1.3.1 Achieve improved Manufacturing Product Quality and Yield via optimized Specs
 - 1.3.2 Support an efficient and consistent TechNode (TN) transfer from TD to HVM sites with critical facilities system alignment to mitigate process variability that may result in product deviations and qualification delay.
 - 1.3.3 Enable easier information sharing across sites
 - 1.3.4 Enable standardized system designs and FAB designs for new Construction
- 1.4 The document is intended as a reference document to supplement each Front-End site's "Quality Specifications". Refer to the specific Quality Specifications published from each Site, for the exact Quality Specifications for that site.
 - 1.4.1 Each Front-End Site has specific "Site Quality Specifications" applicable to the individual Site, that are documented and published.

1.4.2 Each Front-End Site's "Site Quality Specifications" shall take precedence over this Global Front-End Spec Alignment Critical Specification document, for that individual Site.

1.4.3 Each Front-End Site's "Site Quality Specifications", and not the Global Front-End Spec Alignment Critical Specifications, shall be used for the Quality Management of that Site, including Spec Conformance Reporting, Industry/Customer Audits, etc.

1.5 This document is intended to be modified as specification changes occur. The updates made in the latest version is indicated by Financial Year in red font and **specific parameter changes from previous to current version are highlighted in yellow.**

2.0 Scope

2.1 This document covers only Alignment of Critical Facilities Specifications that have been approved together by TD, OCT and GFAC.

2.2 The document is intended as a reference document which supplements the actual Front-End site's "Quality Specifications".

2.3 Sites impacted:

2.3.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron Front-End sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
% RH	Percent Relative Humidity
°C	Degrees Celsius
AMC	Airborne Molecular Contaminants
APC	Airborne Particulate Contaminants
CC	Contamination Control
CFU/L	Colony Forming Unit per Liter
ECN	Engineering Change Notification
FY	Financial Year
FTT	Facilities Technology Transfer
GFAC	Refers to the Global Facilities Team This is the organization consists of Global and Regional teams, which performs both Facilities and Construction related activities.
GFAC-ENG	Refers to the Global Facilities Engineering Team. This is the team focuses on establishing and driving Global Standards and Programs with focus on Engineering, Design, Operations, and Maintenance, to provide standardized solutions with emphasis on safety, performance, sustainability, innovation, reliability, and robustness.
GFAC-REG	Refers to the Global Facilities Regional Team This is the team focuses on alignment of standards to site teams, deploys programs roll-out & penetration to site teams, drive Programs Progress.
MΩ	Megaohm, 1 x 10 ⁶ Ohms

Term/Acronym	Definition / Description
OCT	Operations Central Team
PPB, PPBV	Parts Per Billion, Parts Per Billion by Volume
PPT	Parts per Trillion
QM	Quality Management Team
ST	Site Team comprises Site Facilities, Site QM and Site CC
TD	Technology Development Team
TM	Transfer Manager
TN	Tech Node

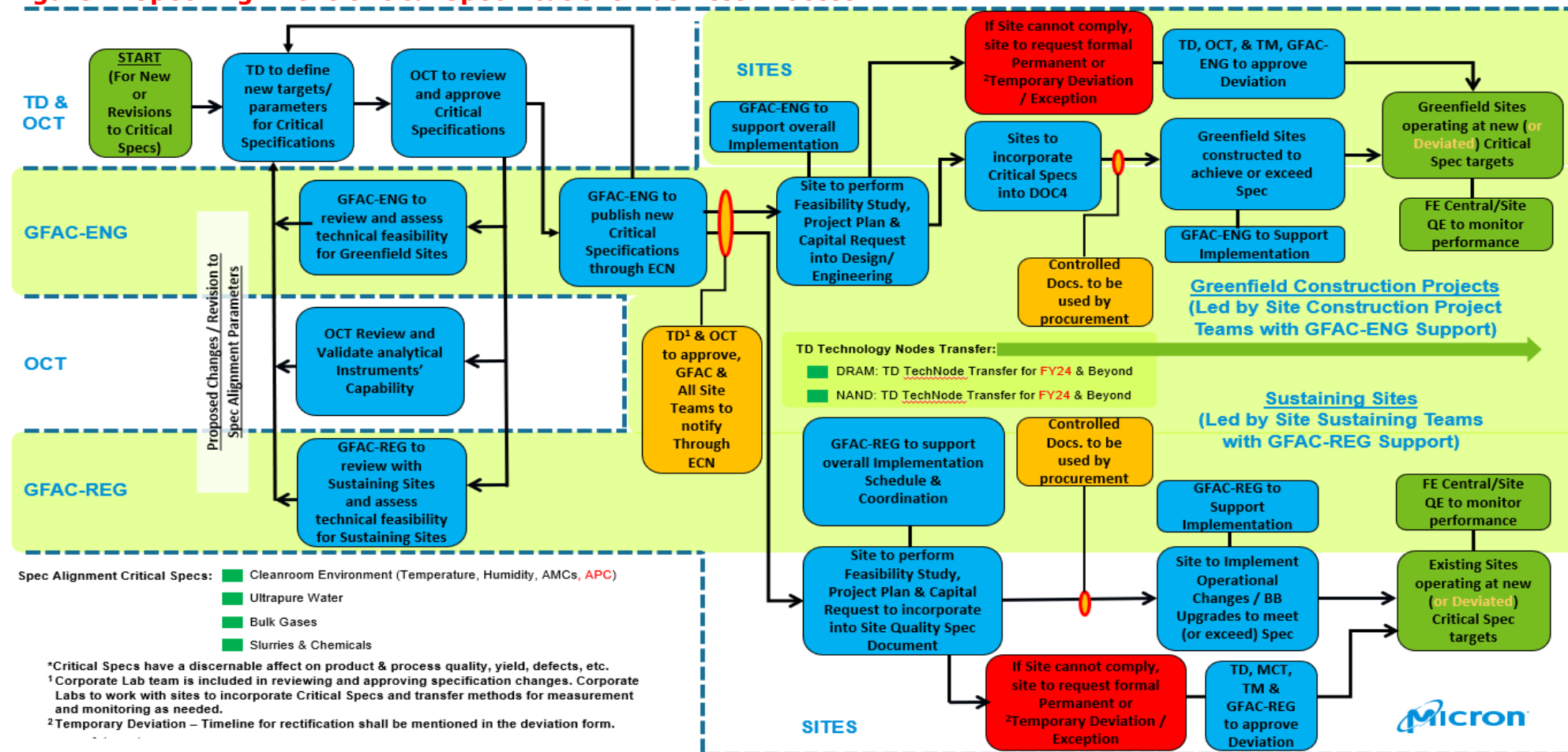
4.0 Roles and Responsibilities

Roles	Responsibilities
OCT and TD	- Develop and approve new Critical Specifications based on production quality/yield benefits and technology node requirements to promote alignment. - Interface with Site Manufacturing and Corporate Labs to adopt the new Critical Specifications. This includes supporting Site risk analysis, project implementation, and post-impact analysis.
GFac- Engineering	- Review, assess technical feasibility, and publish the new Critical Specifications. - Interface with Site Construction Project Teams to adopt the new Critical Specifications into new Greenfield FAB Construction projects. This includes supporting the Engineering, Design, Construction, and project execution. - Introduce and recommend new Facilities system technologies, equipment, and processes that will allow the new Critical Specifications to be achieved for both new Greenfield FAB Construction projects, and for existing Site upgrade projects.
GFac- Regional	- Review and assess technical feasibility the new Critical Specifications. - Interface with Existing Site Facilities to adopt the new Critical Specifications into Sustaining Sites. This includes supporting Site feasibility studies, project planning, Capital funding requests, and project execution.
Corporate Labs	- Interface with OCT and TD to consult on new Critical Specifications. - Interact with GFAC as needed to support stages of Critical Specification Development. - Develop methods and analysis technology to support implementation of Critical Specifications.
Site Facilities/ Site QM/ Site CC	Adopt the new Critical Specifications. This includes implementing Site feasibility studies, project planning, Capital funding requests, and project execution.
Site Manufacturing	Adopt the new Critical Specifications. This includes implementing Site risk analysis, project planning, project execution, and post impact analysis.

5.0 Spec Alignment Business Process:

5.1 The following Business Process shall be followed for development and implementation of new Spec Alignment Critical Specifications: See **Figure 1 and associated IMPORTANT Note 1**:

Figure 1: Spec Alignment Critical Specifications Business Process



IMPORTANT Note 1:

- Critical specifications related to TechNode (TN) Transfers in FY21 (or before release of this document with FY22 updates) shall be aligned to specifications mentioned in Tables of [FY21], [FY21/FY22], [FY21/FY22/FY24] by VME of TN Transfer.
- Older compliance years will drop from tables as all sites move to newer compliance year specifications.

Note: No official TN transfers initiated in FY23

The site must comply to the Tables for the year in which the transfer was initiated and meet compliance by date TN reaches VME as published by the TN milestone schedule.

For example, N160 transfer initiated in FY21, so it must comply with all tables marked by FY21 upon its VME, Tables without FY21 mark are not required to be met until next TN transfer.

Note: Critical Specifications for a site need to be met as follows:

Specification compliance year determined by the TechNode (TN) initiation date based on the date of 1st ECN sent from TD to alpha/HVM site. Spec compliance alignment is TN VME date.

Example Table for Spec Compliance date for current TN Transfers as of date 12/4/23 (take note the VME date may change and adjust Year for compliance)

Please refer to the following Table for Spec Compliance requirement for different TechNodes.

TechNode (TN)	Initiated Transfer Date (Spec Compliance Year)	VME Date	Notes
N160	FY21	6/6/24	Spec changes later than FY21 do not apply to N160
D160	FY22	1/31/25	Apply FY22 when changed from FY21
N170	FY24	3/18/27	

When a later spec compliance year reflects a relaxed specification as compared to previous, the more relaxed specification should be adopted. If the new spec compliance year is more stringent than the previous spec compliance year(s), the node is 'grandfathered' to continue at older compliance year. The next TN to transfer would require meeting the new Spec compliance year, which may be the trigger to change also for older nodes.

- 5.2 **Figure 2** provides the key elements of the of the Business Process of new Spec Alignment Critical Specifications along with the requirements for each element and the element owners/resources to support the elements.

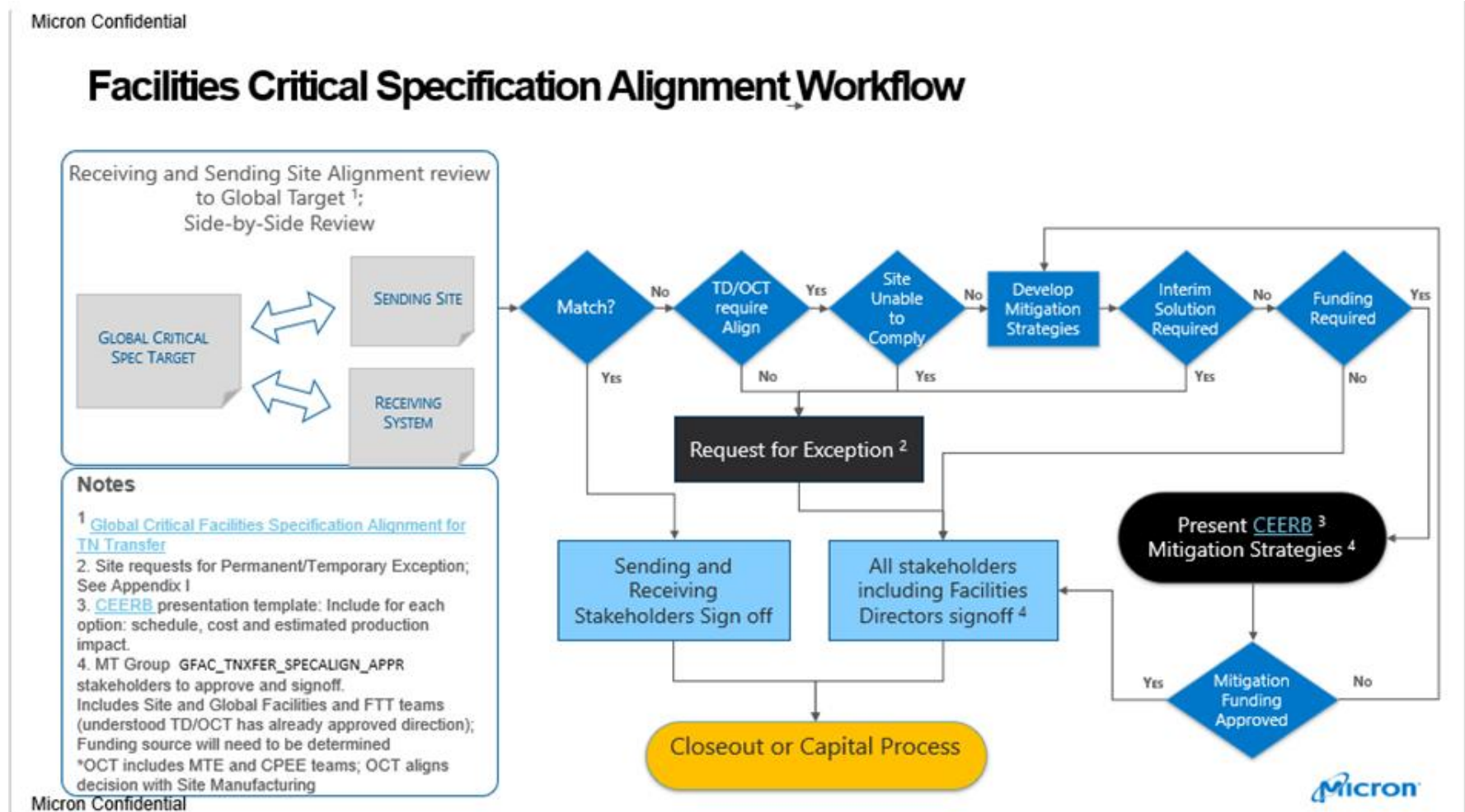
Figure 2: Key elements of the Business Process of new Spec Alignment Critical Specifications

Key Elements of Business Process of Spec Alignment Critical Specification		Owners
Initiation	Alignment of Critical Specifications for new/existing product series of DRAMs and NANDs is critical for manufacturing	TD/FTT
Reviewing	Importance of meeting manufacturing requirements and assessing technical feasibility to achieve required Critical Specifications	OCT, GFAC
Approval	Generated or Revised Critical Specifications of facility systems are critical to meet the manufacturing requirements	OCT, TD/FTT, Site Teams
Publishing	Informing Critical Specifications to sites by ECN is so essential for sites to conform or deviate the required Critical Specifications	GFAC
Deviation	Aligned Critical Specifications may not be followed due to constraints and appropriate formal deviations are necessary. Deviation Approval/Sign-off workflow is provided in Figure3.	Site Teams, TD/FTT, OCT, TM, GFAC
Implementation	Facility systems are to be built and operated to achieve the required Critical Specifications to conform to manufacturing requirements	GFAC, Site Teams, GP
Measurement and Monitoring	Ensure Analytical instruments (on-line, off-line and lab) are capable of measuring all specified parameters and systems are operating within the designed limits	OCT, Corporate Labs, GFAC, Site Teams

BPCSA – Business Process for Critical Specification Alignment

TM refers to Transfer Manager

Note: This document shall strictly be circulated among Micron Team Members only.

5.3 **Figure 3: Approval/Sign-off workflow for Critical Facilities Specification Alignment-Exceptions and Mitigation Strategies**CEERB Link: [CEERB ppt](#)

6.0 Global Facilities System Specification Requirements:

- 6.1 The FAB Cleanroom Environment shall be operated and maintained at the following specifications outlined in **Tables 1 to 6**: Refer to **Appendix II** for additional information on importance of parameter, sampling frequency, sampling size, sampling location(s), analytical instruments and detection limits of analytical instruments.

For the Cleanroom Environment Specs of Backend Assembly and Test (A&T) Sites, refer to the "[Global AT – Cleanroom ISO 8 and Below – Operations, Maintenance, and Controls \(EN\)](#)" document.

Table 1 [FY21-FY24]: Critical Specs for FAB Cleanroom Environment – Temperature & Humidity

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Temperature	22.5 ± 1 °C	Sites: F10, F11, F16 FAB Cleanroom and Reticle Room
Cleanroom Temperature	22.0 ± 1 °C	Sites: F6 FAB Cleanroom and Reticle Room
Cleanroom Humidity	45 ± 5 %RH	Sites: F10 FAB Cleanroom and Reticle Room
Cleanroom Humidity	43 ± 3 %RH	Sites: F4, F6, F11, F16 FAB Cleanroom and Reticle Room

Note:

F4 Temperature is currently not included for Spec Alignment.

F15 Temperature and Humidity is currently not included for Spec Alignment.

Table 2 [FY21-FY24]: Critical Specs for Cleanroom Auxiliary Space Environments – Temperature & Humidity

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Temperature	22.5 ± 2 °C	Sites: F10, F11, F16 Auxiliary Cleanroom Areas - Gowning Room, AMHS Clean Link Bridge
Cleanroom Temperature	22.0 ± 2 °C	Sites: F6 Auxiliary Cleanroom Areas - Gowning Room, AMHS Clean Link Bridge
Cleanroom Humidity	45 ± 5 %RH	Sites: F10 Auxiliary Cleanroom Areas - Gowning Room, AMHS Clean Link Bridge
Cleanroom Humidity	43 ± 3 %RH	Sites: F4, F6, F11, F16 Auxiliary Cleanroom Areas - Gowning Room, AMHS Clean Link Bridge

Note:

F4 Temperature is currently not included for Spec Alignment.

F15 Temperature and Humidity is currently not included for Spec Alignment.

Table 3 [FY21]: Critical Specs for FAB Cleanroom and Auxiliary Space Environment – Particles

Parameter	Specification Requirement	Remarks
Cleanroom Particles	ISO 14644-1 Class 5	FAB Cleanroom
Cleanroom Particles	ISO 14644-1 Class 3	Reticle room, Reticle Cleaning, and Reticle Inspection Area
Cleanroom Particles	ISO 14644-1 Class 6	Gowning Room serving FAB Cleanroom
Cleanroom Particles	ISO 14644-1 Class 6	AMHS Clean Link Bridge
Cleanroom Particles	ISO 14644-1 Class 7	Adjacent Equipment Area/ Wipedown/ Passthrough connected to FAB Cleanroom (i.e. FAB Decon Room)

Table 3 [FY22-FY24]: Critical Specs for FAB Cleanroom and Auxiliary Space Environment – Particles

Parameter	Specification Requirement	Remarks
Cleanroom Particles	ISO 14644-1 Class 5	FAB Cleanroom
Cleanroom Particles	ISO 14644-1 Class 3	Reticle room, Reticle Cleaning, and Reticle Inspection Area
Cleanroom Particles	ISO 14644-1 Class 3	EUV reticle room (Note: Spec is ISO Class 3, performance preferred to operate exceed ISO Class 3)
Cleanroom Particles	ISO 14644-1 Class 5	EUV Cleanroom
Cleanroom Particles	ISO 14644-1 Class 6	Gowning Room serving FAB Cleanroom
Cleanroom Particles	ISO 14644-1 Class 6	AMHS Clean Link Bridge
Cleanroom Particles	ISO 14644-1 Class 7	Adjacent Equipment Area/ Wipedown/ Passthrough connected to FAB Cleanroom (i.e. FAB Decon Room)

Table 4 [FY21-FY24]: Critical Specs for Probe Cleanroom and Auxiliary Space Environment – Particles

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Particles	ISO 14644-1 Class 7	Probe Area (or Probe tool minienvironment)
Cleanroom Particles	ISO 14644-1 Class 8	Gowning Room serving Probe Area
Cleanroom Particles	ISO 14644-1 Class 8	Adjacent Equipment Area/ Wipedown/ Passthrough connected to Probe Area (i.e. Probe Decon Room)

Note: If probe tool mini-environment requirement is used, unclassified probe room doesn't require Gowning room and Decon area classification.

Table 5 [FY21-FY24]: Critical Specs for FAB Cleanroom Environment (Photo Area) – AMCs

Parameter	Specification Requirement	Remarks
Acids, Total	< 5 ppbv	Photo Area
Ammonia/Amines, Total	< 4 ppbv	Photo Area
Condensables	< 50 ppbv	Photo Area
Refractories	< 2.5 ppbv	Photo Area

Note: Ammonia is added. Spec for refractories increased to 2.5ppbv as ASML warranty requirement is stated in a separate Table 5b.

Table 5a [FY21-FY24]: Critical Specs for FAB Cleanroom Environment (Photo Area) – AMCs For Sites that have AMC Monitoring capability for Acid speciation, then the following Acid Specs shall apply:

Parameter	Specification Requirement	Remarks
Acids		Photo Area – Acids, based on speciation
- Fluoride	< 5 ppbv	
- Chloride	< 2 ppbv	
- Nitrite	< 10 ppbv	
- Nitrate	< 1 ppbv	
- Sulfate	< 1 ppbv	

Tables 5b: [FY21-FY24] Critical Specs for ASML Warranty – AMCs

Parameter	Specification Requirement	Remarks
Acids, Total	< 5 ppbv	ASML ACC Air Intake
Ammonia/Amines, Total	< 25 ppbv	ASML ACC Air Intake
Condensables	< 50 ppbv	ASML ACC Air Intake
Refractories	< 0.1 ppbv	ASML ACC Air Intake

Note: Table 5b is created with ASML Warranty requirement.

Refer to attachment for the required set-up needed to achieve critical specifications for ASML Warranty.

Table 6 [FY21-FY24]: Critical Specs for FAB Cleanroom Environment - AMCs

Parameter	Specification Requirement	Remarks
Acids, Total	< 10 ppbv	All Other Areas except Photo
Ammonia/Amines, Total	< 1.5 ppbv	Reticle Area
Ammonia/Amines, Total	< 30 ppbv	All Other Areas except Reticle and Photo (Note: If CMP Area is segregated, then spec is <50 ppbv instead of 30 ppbv)
Ozone (O3)	< 5 ppbv	SOD Process
Hydrogen Sulfide (H2S)	< 5 ppbv	Copper Utility Zone. The area immediately surrounding Copper Plating tools and Copper CMP processing tools.

Table 6a [FY21-FY24]: Critical Specs for FAB Cleanroom Environment - AMCs For Sites that have AMC Monitoring capability for Acid speciation, then the following Acid Specs shall apply:

Parameter	Specification Requirement	Remarks
Acids		All Other Areas except Photo – Acids, based on speciation
- Fluoride	< 5 ppbv	
- Chloride	< 2 ppbv	
- Nitrite	< 10 ppbv	
- Nitrate	< 1 ppbv	
- Sulfate	< 1 ppbv	

- 6.2 The Ultrapure Water and Hot Ultrapure Water Systems shall be operated and maintained at the following specifications outlined in **Tables 7 [FY21-FY24] and Table 8 [FY21/FY22]**: Refer Appendix III for additional information on importance of parameter, sampling method, sampling location, frequency, best in class analytical instruments, monitoring analytical instruments and detection limits of analytical instruments.

Table 7a [FY21-FY24]: Critical Specs for UPW

Parameter	Specification Requirement	Remarks
Temperature	23 ± 0.5 C	
Resistivity	≥ 18.2 MΩ-cm	
Maximum Particle Count	< 0.3 counts/mL @ ≥ 0.05µm size	
Maximum Dissolved Silica	< 50 ppt	
Maximum Total Silica	< 50 ppt	
Maximum Dissolved Oxygen (DO)	< 2 ppb	
Maximum Dissolved Nitrogen	14 ± 1 ppm 9 ± 1 ppm	Fab4 maintain at 14 ± 1 ppm. F10N, F10X and F10A maintain at 9 ± 1 ppm.
Maximum TOC	< 1 ppb	
Boron	< 50 ppt	
H2O2	< 10 (ppb)	
Bacteria	< 1 CFU/L	
Na+ Cation	< 2 ppt	
K+ Cation	< 5 ppt	
Ammonium	< 10 ppt	
Ca++ Cations	< 2 ppt	
Mg++ Cations	< 1 ppt	
Cu++ Cations	< 1 ppt	
Fe++ Cations	< 2 ppt	
Mn++ Cations	< 1 ppt	

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Parameter	Specification Requirement	Remarks
Ni++ Cations	< 2 ppt	
Zn++ Cations	< 2 ppt	
Tetra Methyl Ammonium (TMA+)	< 5 ppt	Clarifying parameter name
Al+++ Cations	< 1 ppt	
Sr	< 1 ppt	
Ba	< 1 ppt	
Cr	< 1 ppt	
Sn	< 1 ppt	
Li	< 1 ppt	
Ag	< 1 ppt	
Pb	< 1 ppt	
Mo	< 1 ppt	
W	< 1 ppt	
Cd	< 1 ppt	
Co	< 1 ppt	
Ti	< 1 ppt	
V	< 1 ppt	
Fluoride	< 10 ppt	
Chloride	< 5 ppt	
Bromide	< 5 ppt	
Nitrate	< 5 ppt	
Nitrite	< 5 ppt	
Phosphate	< 10 ppt	
Sulphate	< 10 ppt	

Table 7b [FY24]: Critical Specs for UPW

Parameter	Specification Requirement	Remarks
Trimethylamine (TMA)	< 5 ppt	Added in FY24

Note: Follow Table 7a and 7b

Table 8 [FY21-FY24]: Critical Specs for HUPW

Parameter	Specification Requirement	Remarks
Temperature	65 ± 1.0 C	
Resistivity	≥ 18.2 MΩ-cm	
Maximum Particle Count	< 0.3 counts/mL @ ≥ 0.05µm size	
Maximum Dissolved Silica	< 50 ppt	
Maximum Total Silica	< 50 ppt	
Maximum TOC	< 1 ppb	
F-	< 10 ppt	

Note: HUPW critical specification shall match UPW critical specification except for temperature

The Bulk Gases shall be operated and maintained at the following specifications as outlined in **Tables 9 through 17**.

Table 9 [FY21-FY24]: Specifications for PN₂

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 105 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
O ₂	≤ 10 ppb	Maximum impurity concentration as measured at the outlet of the purifier or gas plant (point of location in which the specification is met).
H ₂ O	≤ 10 ppb	
CO	≤ 10 ppb	
CO ₂	≤ 10 ppb	

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Parameter	Specification Requirement	Notes/Comments
H ₂	≤ 10 ppb	
THC (expressed as CH ₄)	≤ 10 ppb	
Particles	Max: 5 particles/scf @ ≥ 0.1 μm Avg: 1 particle/scf @ ≥ 0.1 μm	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas Or By Pressurized Gas Cylinder" for test method.

Table 10 [FY21-FY24]: Specifications for GN2

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 105 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
O ₂	≤ 100 ppb	Maximum impurity concentration as measured at the outlet of the purifier or gas plant (point of location in which the specification is met).
H ₂ O	≤ 100 ppb	
CO	≤ 100 ppb	
CO ₂	≤ 100 ppb	
H ₂	≤ 100 ppb	
THC (expressed as CH ₄)	≤ 100 ppb	

Table 11 [FY21-FY24]: Specifications for UN₂

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 105 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
O ₂	$\leq 2,000$ ppb	Maximum impurity concentration as guaranteed by COC provided by gas supplier.
H ₂ O	$\leq 2,000$ ppb	
H ₂	$\leq 2,000$ ppb	

Table 12 [FY21-FY24]: Specifications for PO₂

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 65 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
Ar	≤ 50 ppb	Maximum impurity concentration as measured at the outlet of the purifier or location downstream of gas yard in which specification is met.
N ₂	≤ 10 ppb	
H ₂ O	≤ 10 ppb	
CO	≤ 10 ppb	
CO ₂	≤ 10 ppb	
H ₂	≤ 10 ppb	
THC (expressed as CH ₄)	≤ 10 ppb	
Particles	Max: 5 particles/scf @ ≥ 0.1 μ m Avg: 1 particle/scf @ ≥ 0.1 μ m	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas Or By Pressurized Gas Cylinder" for test method.

Table 13 [FY21-FY24]: Specifications for PH₂

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 65 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
N ₂	≤ 10 ppb	Maximum impurity concentration as measured at the outlet of the purifier or location downstream of gas yard in which specification is met.
O ₂	≤ 10 ppb	
H ₂ O	≤ 10 ppb	
CO	≤ 10 ppb	
CO ₂	≤ 10 ppb	
THC (expressed as CH ₄)	≤ 10 ppb	
Particles	Max: 5 particles/scf @ ≥ 0.1 μm Avg: 1 particle/scf @ ≥ 0.1 μm	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas Or By Pressurized Gas Cylinder" for test method.

Table 14 [FY21-FY24]: Specifications for PAr

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 65 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
N ₂	≤ 10 ppb	Maximum impurity concentration as measured at the outlet of the purifier or location downstream of gas yard in which specification is met.
O ₂	≤ 10 ppb	
H ₂ O	≤ 10 ppb	
CO	≤ 10 ppb	
CO ₂	≤ 10 ppb	
H ₂	≤ 10 ppb	

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Parameter	Specification Requirement	Notes/Comments
THC (expressed as CH ₄)	≤ 10 ppb	
Particles	Max: 5 particles/scf @ > 0.1 μm Avg: 1 particle/scf @ > 0.1 μm	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas Or By Pressurized Gas Cylinder" for test method.

Table 15 [FY21-FY24]: Specifications for PHe

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 65 psig	Minimum pressure as measured at end of line. Site must evaluate individual tool requirements as well as compatibility to existing distribution system and adjust pressure requirement accordingly.
O ₂	≤ 10 ppb	Maximum impurity concentration as measured at the outlet of the purifier or location downstream of gas yard in which specification is met.
H ₂ O	≤ 10 ppb	
CO	≤ 10 ppb	
CO ₂	≤ 10 ppb	
H ₂	≤ 10 ppb	
THC (expressed as CH ₄)	≤ 10 ppb	
Particles	Max: 5 particles/scf @ ≥ 0.1 μm Avg: 1 particle/scf @ ≥ 0.1 μm	Refer to SEMI C97-1118 "Specification For Determination of Particle Levels of Gases Delivered as Pipeline Gas Or By Pressurized Gas Cylinder" for test method.

Table 16 [FY22-FY24]: Specifications for High Pressure CO₂ (Wets & Mask Process Areas)

Parameter	Specification Requirement	Notes/Comments
Pressure	800 ± 60 psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust pressure requirement accordingly.
H ₂ O	≤ 10 ppbv	Maximum impurity as measured post purifier.
CO	≤ 10,000 ppbv	
Total S	≤ 10,000 ppbv	
Sum Others	≤ 29,900 ppbv	
CO ₂ Purity	≥ 99.995 ppbv	Minimum CO ₂ purity as measured post purifier

Entegris PS32-MGTxx-CO₂ or equivalent purifier recommended to meet outlet specifications.

Note to Procurement: Procurement to request gas supplier to provide a quarterly Certificate of Compliance with impurities as shown in **Appendix V**.

Table 17 [FY22-FY24]: Specifications for CO₂ (Photo, CVD, Dry Etch Process Areas)

Parameter	Specification Requirement	Notes/Comments
Pressure	≥ 70 psig	Minimum pressure as measured at the point of use as for system design specification. Site to evaluate individual tool requirements as well as existing distribution system and adjust pressure requirement accordingly.
H ₂ O	≤ 1 ppbv	Maximum impurity as measured post purifier.
O ₂	≤ 1 ppbv	

Parameter	Specification Requirement	Notes/Comments
N ₂	<30 ppmv	
CO	≤ 1 ppbv	
THC (as CH ₄)	<20ppmv	
Total S (COS, CS ₂ , H ₂ S)	≤ 1 ppbv	
Acetaldehyde	<0.050 ppmv	
Benzene	≤ 0.001 ppbv	
NH ₃	≤ 0.005 ppbv	
NO	≤ 1 ppbv	
NO ₂	≤ 1 ppbv	
CO ₂ Purity	99.995%	Minimum CO ₂ purity as measured post purifier
Particles	≤ 10 particles/m ³	For particle size ≥ 0.1 µm (ISO Class 1 > ISO 14644-1 Class 1)
	≤ 2 particles/m ³	or particle size ≥ 0.2 µm (ISO Class 1 > ISO 14644-1 Class 1)

Note 1:

The table is applicable to Tech Nodes from FY22 to FY24. Latest changes in parameters are highlighted in yellow.

Regarding Conversion Qualification Requirement, the current LCO₂ users will not require qualification when new specification is lower than current site LCO₂ contract specification. CO₂ users converting from spec gas cylinder to Bulk LCO₂ will require qualification when bulk spec is more relaxed than current spec gas spec.

Entegris PS32-MGTxx-CO₂ or equivalent purifier recommended to meet outlet specifications.

Note 2 (For Procurement): *Procurement team to request gas supplier to provide a quarterly Certificate of Compliance with impurities as shown in **Appendix V**.*

6.3 Blended chemical and slurry shall be operated and maintained at the following specifications as outlined in **Table 18a [FY22] and 18b [FY24]**

Note: Sites to continue using the specifications in **Table 18a** until new Tech Nodes are introduced, for which **Table 18b** will be used.

Table 18a [FY22]: Specifications for Blended Chemical and Slurry

S/N	Blended Chemical / Slurry	Blend Details			Conductivity (Based on Lab Data)			Assay H2O2 (Based on Lab Data)			Solid Content (Based on Lab Data)			pH (Based on Lab Data)		
		Target Site (Transfer site)	Blend Material	Blend Ratio by Weight	Target site LCL (Us/cm)	Target site UCL (Us/cm)	Target CL (Us/cm)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL	Target site UCL	Target CL
1	D9228	F4	UPW: D9228	6.5: 1	46.7	53.22	49.96	NA	NA	NA	1.94	2.3	2.12	4.6	4.94	4.77
2	CSL9044C	F4	UPW: 9044: H2O2(31%)	9.42: 1.055: 0.42 (8.93: 1: 0.4)	233.91	264.69	249.3	1.05	1.19	1.12	1.08	1.6	1.34	7.0	7.55	7.16
3	HS8800-C2	F4	UPW: HS8800	1: 1.08	141.29	191.71	166.5	NA	NA	NA	1.98	2.16	2.07	8.41	9.23	8.82
4	W7300-B21	F4	W7300: H2O2 (31%)	14.7: 1 (1: 0.068)	2977	3285	3131	1.889	2.049	1.969	3.4	3.842	3.621	2.19	2.35	2.27
5	KP3100 + STI2910	F4	UPW: KP3100: STI2910	100: 1: 1	708.69	748.31	728.5	NA	NA	NA	NA	NA	NA	4.11	4.31	4.21
6	0.063 TMAH	F4	UPW: TMAH 2.38%	38: 1.015 (37.4: 1)	1400	1600	1500	0.061	0.065	0.063	NA	NA	NA	NA	NA	NA
7	STI2910	F4	UPW: STI2910	311.5: 1.05 (297: 1)	100.9	107.8	104.35	NA	NA	NA	NA	NA	NA	4.14	4.38	4.26
8	STI2401	F4	UPW: STI2401	3: 1.1 (2.72: 1)	797.5	882.5	840	NA	NA	NA	2.596	2.876	2.736	6.19	6.43	6.31
9	Glanzox 3105	F4	UPW: Glanzox 3105	19: 1.05 (18.1: 1)	153.13	313.13	233.13	NA	NA	NA	0.41	0.61	0.51	9.68	10.32	10.0

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S/N	Blended Chemical / Slurry	Blend Details			Conductivity (Based on Lab Data)			Assay H2O2 (Based on Lab Data)			Solid Content (Based on Lab Data)			pH (Based on Lab Data)		
		Target Site (Transfer site)	Blend Material	Blend Ratio by Weight	Target site LCL (Us/cm)	Target site UCL (Us/cm)	Target CL (Us/cm)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL	Target site UCL	Target CL
10	Dilute MO2 (0.2%)	F10NXA	UPW: MO2 (5%)	25: 1	3660	3560	3610	0.188	0.204	0.197	NA	NA	NA	NA	NA	NA
11	PL8105	F4	UPW: 8105: H2O2(31%)	1.6: 1.1: 0.27	7255	8355	7805	2.184	2.784	2.484	7.15	8.41	7.78	9.48	9.72	9.6
12	HF100 (0.5%)	F4	UPW: HF49%	92.1: 1	7290	7510	7400	0.47	0.53	0.50	NA	NA	NA	NA	NA	NA
13	HSP 700-9	F4	UPW: HSP700-9	11: 1.075	6540	7188	6864	NA	NA	NA	NA	NA	NA	NA	NA	NA
14	PL6501	F4	UPW: PL6501	10: 1	11.57	26.77	19.17	NA	NA	NA	NA	NA	NA	NA	NA	NA
15	PL6508	F4	UPW: PL6508	37: 1.01	305.5	370.5	338	NA	NA	NA	NA	NA	NA	NA	NA	NA
16	W8503	F4	UPW: W8503: H2O2	3: 1: 0.056	893	1093	993	0.453	0.607	0.53	Nil	Nil	Nil	NA	NA	NA
17	PL6116	F10NXA	UPW: PL6116	5: 1	1700	2700	2200	NA	NA	NA	2.3	3.3	2.8	2.5	3.5	3.0
18	DEV 4262	F16/F11	UPW: OPD25C: OPD4000SC	9.48: 1: 0.0015	NA	NA	NA	2.380	2.390	2.385	NA	NA	NA	NA	NA	NA
19	SRS2233	F10NX	UPW: SRS2233	1.1: 1	530	570	550	NA	NA	NA	4.267	7.697	5.982	NA	NA	NA
20	CES330XD1	F4	UPW: CES330XD1	12: 1.05	250	310	280	NA	NA	NA	0.72	0.90	0.81	3	3.4	3.2
21	FPM (HF+ H2O2)	F16	UPW: HF49% + UPW: H2O2 31%	4.48: 0.057 + 29.9: 1	NA	NA	NA	0.52	0.68	0.60	NA	NA	NA	NA	NA	NA
22	CES330XD4	F4	UPW: CES330XD4	4: 1.05	20	26	23	NA	NA	NA	1.8	2.2	2	4	5	4.5

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S/N	Blended Chemical / Slurry	Blend Details			Conductivity (Based on Lab Data)			Assay H2O2 (Based on Lab Data)			Solid Content (Based on Lab Data)			pH (Based on Lab Data)		
		Target Site (Transfer site)	Blend Material	Blend Ratio by Weight	Target site LCL (Us/cm)	Target site UCL (Us/cm)	Target CL (Us/cm)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL	Target site UCL	Target CL
23	NP6501	F10N	UPW: NP6501	20: 1	280	350	315	NA	NA	NA	1.81	2.1	1.96	9.8	10.5	10.2
24	PL6118	F4	UPW: PL6118	5: 1.01	765	855	810	NA	NA	NA	0.32	0.44	0.38	2.64	2.78	2.71
25	PL6125N	F4	UPW: PL6125	3: 1.06	5470	6510	5990	NA	NA	NA	NA	NA	NA	1.75	1.95	1.85
26	STI2601	F10XA	UPW: STI2601	2.98: 1	750	850	800	NA	NA	NA	2.5	2.8	2.65	6.05	6.35	6.2
27	W8900	F4	UPW: W8900: H2O2	5: 1: 0.58	605	745	675	2.9	3.1	3.0	0.15	0.21	0.18	2.5	2.9	2.7
28	SRS5001	F4	UPW: SRS5001	1: 1	222.4	272.4	247.4	NA	NA	NA	NA	NA	NA	3.11	3.41	3.26
29	SRS2298	F10X	UPW: SRS2298	1: 1	240	620	430	NA	NA	NA	NA	NA	NA	5.0	6.0	5.5

Table 18b [FY24]: Specifications for Blended Chemical and Slurry

S/N	Blended Chemical / Slurry	Blend Details			Conductivity (Based on Lab Data)			Assay H2O2 (Based on Lab Data)			Solid Content (Based on Lab Data)			pH (Based on Lab Data)		
		Target Site (Transfer site)	Blend Material	Blend Ratio by Weight	Target site LCL (Us/cm)	Target site UCL (Us/cm)	Target CL (Us/cm)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL	Target site UCL	Target CL
1	D9228	F4	UPW: D9228	6.5: 1	46.7	53.22	49.96	NA	NA	NA	1.94	2.3	2.12	4.6	4.94	4.77
2	CSL9044C	F4	UPW: 9044: H2O2(31%)	8.57:1.0:0.40	233.91	264.69	249.3	1.05	1.19	1.12	1.08	1.6	1.34	7.0	7.55	7.16
3	HS8800-C2	F4	UPW: HS8800	1: 1.08	141.29	191.71	166.5	NA	NA	NA	1.98	2.16	2.07	8.41	9.23	8.82
4	W7300-B21	F4	W7300: H2O2 (31%)	14.7: 1 (1: 0.068)	2977	3285	3131	1.84	2.02	1.93	3.4	3.8	3.6	2.16	2.36	2.26
5	KP3100 + STI2910	F4	UPW: KP3100: STI2910	100: 1: 1	708.69	748.31	728.5	NA	NA	NA	NA	NA	NA	4.11	4.31	4.21
6	0.063 TMAH	F4	UPW: TMAH 2.38%	38: 1.015 (37.4: 1)	1400	1600	1500	0.061	0.065	0.063	NA	NA	NA	NA	NA	NA
7	STI2910	F4	UPW: STI2910	311.5: 1.05 (297: 1)	100.9	107.8	104.35	NA	NA	NA	NA	NA	NA	4.14	4.38	4.26
8	STI2401	F4	UPW: STI2401	3: 1.1 (2.72: 1)	799	900	849	NA	NA	NA	2.6	2.9	2.8	6.08	6.45	6.27
9	Glanzox 3105	F4	UPW: Glanzox 3105	19: 1.05 (18.1: 1)	103	362	233	NA	NA	NA	0.49	0.57	0.53	9.48	10.28	9.88
10	Dilute MO2 (0.2%)	F10NXA	UPW: MO2 (5%)	25.5:1	4890	5190	5040	0.188	0.204	0.197	NA	NA	NA	NA	NA	NA
11	PL8105	F4	UPW: 8105: H2O2(31%)	1.17 : 1.0 : 0.224	7430	8860	8140	2.184	2.784	2.484	7.16	8.41	7.78	9.48	9.73	9.6
12	HF100 (0.5%)	F4	UPW: HF49%	92.1: 1	7290	7510	7400	0.47	0.53	0.50	NA	NA	NA	NA	NA	NA

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S/N	Blended Chemical / Slurry	Blend Details			Conductivity (Based on Lab Data)			Assay H2O2 (Based on Lab Data)			Solid Content (Based on Lab Data)			pH (Based on Lab Data)		
		Target Site (Transfer site)	Blend Material	Blend Ratio by Weight	Target site LCL (Us/cm)	Target site UCL (Us/cm)	Target CL (Us/cm)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL	Target site UCL	Target CL
13	HSP 700-9	F4	UPW: HSP700-9	11: 1.075	6540	7188	6864	NA	NA	NA	NA	NA	NA	NA	NA	NA
14	PL6501	F4	UPW: PL6501	10: 1	13.1	24.7	18.9	NA	NA	NA	NA	NA	NA	8.97	10.08	9.52
15	PL6508	F4	UPW: PL6508	37: 1.01	322	363	343	NA	NA	NA	NA	NA	NA	3.22	3.48	3.35
16	W8503	F4	UPW: W8503: H2O2	3: 1: 0.056	910	1060	989	0.46	0.57	0.51	0.29	0.38	0.33	2.49	2.81	2.65
17	PL6116	F10NXA	UPW: PL6116	5: 1	1700	2700	2200	NA	NA	NA	2.3	3.3	2.8	2.5	3.5	3.0
18	DEV 4262	F16/F11	UPW: OPD25C: OPD4000SC	9.48: 1: 0.0015	NA	NA	NA	2.380	2.390	2.385	NA	NA	NA	NA	NA	NA
19	SRS2233	F10NX	UPW: SRS2233	1.1: 1	530	570	550	NA	NA	NA	4.267	7.697	5.982	NA	NA	NA
20	CES330XD1	F4	UPW: CES330XD1	12.3:1.0	250	310	280	NA	NA	NA	0.72	0.90	0.81	3	3.4	3.2
21	FPM (HF+ H2O2)	F16	UPW: HF49% + UPW: H2O2 31%	4.48: 0.057 + 29.9: 1	NA	NA	NA	0.52	0.68	0.60	NA	NA	NA	NA	NA	NA
22	CES330XD4	F4	UPW: CES330XD4	4: 1.05	20	26	23	NA	NA	NA	1.8	2.2	2	4	5	4.5
23	NP6501	F10N	UPW: NP6501	20: 1	280	350	315	NA	NA	NA	1.81	2.1	1.96	9.8	10.5	10.2
24	PL6118	F4	UPW: PL6118	5: 1.01	765	855	810	NA	NA	NA	0.32	0.44	0.38	2.64	2.78	2.71
25	PL6125N	F4	UPW: PL6125	3: 1.06	5470	6510	5990	NA	NA	NA	NA	NA	NA	1.75	1.95	1.85
26	STI2601	F10XA	UPW: STI2601	2.98: 1	750	850	800	NA	NA	NA	2.5	2.8	2.65	6.05	6.35	6.2

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S/N	Blended Chemical / Slurry	Blend Details			Conductivity (Based on Lab Data)			Assay H2O2 (Based on Lab Data)			Solid Content (Based on Lab Data)			pH (Based on Lab Data)		
		Target Site (Transfer site)	Blend Material	Blend Ratio by Weight	Target site LCL (Us/cm)	Target site UCL (Us/cm)	Target CL (Us/cm)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL (wt%)	Target site UCL (wt%)	Target CL (wt%)	Target site LCL	Target site UCL	Target CL
27	W8900-V01	F4	UPW: W8900: H2O2	5: 1: 0.58	605	745	675	2.9	3.1	3.0	0.15	0.21	0.18	2.5	2.9	2.7
28	SRS2298	F10X	UPW: SRS2298	1: 1	240	620	430	NA	NA	NA	NA	NA	NA	5.0	6.0	5.5
29	W8900-V16	F11	UPW:W8900:H2O2	1:5:0.21	775	895	845	0.98	1.08	1.04	0.11	0.25	0.18	2.5	2.8	2.7
30	W7279-V03	F10	W7279-V03:H2O2:H2O2	1.8:1:0.2	258	352	300	1.78	2.25	2	3.9	4.74	4.32	3.85	4.54	4.2
31	JSR CMS-AC010	F15	UPW: AC010	37.9:1	3600	4400	4000	NA	NA	NA	NA	NA	NA	12.45	12.81	12.63
32	PL5111	F15	UPW:PL5111:H2O2	0.9:1.0:0.1	5207	5374	5291	NA	NA	NA	NA	NA	NA			
33	TMAH 2.38%+S(DEV 4262)	F15-B2	UPW: 20% TMAH (OPD20C): OPD4000SC (surfactant)	7.4:1:0.0013	NA	NA	NA	TMAH 2.38	TMAH 2.39	TMAH 2.385	NA	NA	NA	NA	NA	NA
34	NH4OH 2%	F15	UPW: NH4OH 28%	48:4	NA	NA	NA	NH4OH 1.82	NH4OH 2.22	NH4OH 2.02	NA	NA	NA	NA	NA	NA

Note: Additional blended chemical / slurry data will be added as new material is introduced

7.0 Document Control

7.1 Approval: GFAC_TNXFER_SPECALIGN_APPR

7.2 Notification:

GFAC_TNXFER_SPECALIGN_NOTIFY
GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

7.3 Final Viewer Access

All Micron Team Members at all Sites

7.4 Retention

Adherence to the requirements specified in the
[**Global Facilities – Document Control Procedure**](#) and
[**Global Quality Laboratory Management Standard**](#)

7.5 Review

Annually

8.0 Revision History

Revision #	Section	Details of Changes	Date
0	NA	New document	12/08/20
1	Whole document	Added corresponding Tables to differentiate specifications between FY21, FY22 and both (i.e FY21/FY22)	09/30/21
	6.1	Added a new section on Approval/Sign-off workflow for Deviation from Critical Specification <i>Table 3: Added EUV Area requirement</i>	
	6.3	<i>Table 5b: Added new table on Critical Specs for ASML Warranty – AMCs to separate Photo area and ASML air intake in AMC specification</i>	
	6.4	Addition of Table 16 and Table 17 for Bulk CO ₂ Specification	
	Appendix II	Addition of Table 18 for specifications for blended chemicals and slurries	
	Appendix IV	Additional information under Future Instrument and Sampling location	
	Appendix V	Added appendix on Required Set-Up for Achieving Critical Specifications for ASML Warranty Added appendix on Sample Certificate of Compliance with impurities	
2	6.1	Added document link for Backend Assembly and Test (A&T) Cleanroom Environment Specs. Removed F15 from being included for Cleanroom Humidity Spec Alignment, and Cleanroom Auxiliary Space Temperature & Humidity Spec Alignment.	02/16/22
3	1.5	Updated to the following: This document is intended to be modified as specification changes occur. The updates made in the latest version is indicated by Financial Year in red font and specific parameter changes from previous to current version are highlighted in yellow.	03/11/24
	3.0	New acronyms GFAC, GFAC-ENG, GFAC-REG, TN	

		Removed the following: FCT, GFC, GFTT, Global Facilities Central Team, Global Facilities Construction Team, Global Facilities Operations Support Team and Global Facilities Planning Team	
	5.1	Added Important Note 1 Figure 1: Updated to replace GFTT/FCT with corresponding Team acronyms (GFac-Reg / GFac-Eng)	
	5.2	Figure 2: Updated to replace GFTT/FCT with corresponding Team acronyms (GFac)	
	5.3	Figure 3: Updated the workflow	
	6.0	Section 6.1: - Combined Table 5 [FY21] and Table 5 [FY22-FY24] into Table 5 [FY21-FY24] - Combined Table 6 [FY21] and Table 6 [FY22-FY24] into Table 5 [FY21-FY24]. Added Ammonia Section 6.2: Table 17 - Updated parameters for N2, THC, Acetaldehyde and CO2 Purity. See item in Yellow Section 6.2: - Added Note under Table 17. See item in Red - Added Table 7b [FY24]: Critical Specs for UPW Section 6.3: - Added a Note. Refer to item in red - Renamed Table 18 to Table 18a [FY22] - Added Table 18b [FY24] - Removed row for SRS5001 - Added rows for W8900-V16/ W7279-V03/ JSR CMS-AC010/ PL5111/ TMAH 2.38% +S(DEV 4262)/ NH4OH 2% - See items in Yellow - Added a Note to Table 18b [FY24]	
	Appendix II	Appendix II: Removed Future Instrument from the Table in Appendix II	
	Appendix III	Appendix III: - Changed title to "Recommended Analytical Instrumentation for UPW Characterization" to Title for Appendix III - Removed columns i) SPC Parameter Type (C/NC/E), Sample Location (Locations), iii) Spec Sample method and iv) Spec Sample Frequency & Duration from the Table in the Appendix	
	Appendix IV	Appendix IV - Added entire appendix	

Appendix I - Site Specific Deviation List

Site: Fab

Technology Node: DRAM/NAND 140S

Review Date: _____ Facility Requesting Deviation: _____ Technology Risk: (Technology/Material/Process/Timeline) Areas: OCT Manager: TD Manager: Transfer Manager: Site Process/CC Manager: Site Facility Manager: FCT/GFTT Manager:	Deviation Description:																				
Impacted Process Area and Workstations:																					
<table><tr><th>Impacted Process Area</th><th>Workstations</th><th>Impact Description</th></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr></table>		Impacted Process Area	Workstations	Impact Description																	
Impacted Process Area	Workstations	Impact Description																			
Justification Summary by site team: (Includes links to data and backup slides as necessary)	Long Term Alignment Direction (include High Level Cost and Schedule Estimate):																				
Mitigation Plan by site team:	Critical Lessons Learnt if Applicable:																				
Approval:																					
<table><tr><th>Department</th><th>Position</th><th>Name</th><th>Date</th></tr><tr><td> </td><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td><td> </td></tr></table>		Department	Position	Name	Date																
Department	Position	Name	Date																		

Note: Any site deviation from the given specification shall be submitted for approval. If a temporary deviation/exception is requested, the time needed for rectification shall be mentioned in the relevant box(es).

Appendix II – Additional Information on Importance of CR/AMC Parameters, Sampling, Analysis and Monitoring (for reference)

Environment/ Contamination	Parameter	UOM	Analytical Instrumentation	Analytical Detection Limit (DL)	Fab Sampling Location	Sample Frequency & Duration (minimum)
Clean Room Environment (fab)	Cleanliness Classification	Particle/m ³	PMS / Lighthouse	0.3 um – Online	30*30m	Continuous
Clean Room Environment (fab)	Cleanliness Classification	Particle/m ³	PMS / Lighthouse	0.1 um – Offline	vA (A: Fab area, m2)	Monthly (if no online measurement) or Yearly (if online measurement)
Airborne Molecular Contamination	Total Acids	ppbv	PMS Air Sentry II	0.1 ppbv	35x35m (Fab Level) 50x50m (Photo Sub Fab)	3 points/hr
Airborne Molecular Contamination	Acids Speciation	ppbv	NAVI EM201/MAX-7 SOT-100 series	Based on species	35x35m (Fab Level) 50x50m (Photo Sub Fab)	1 point/12hr
Airborne Molecular Contamination	Ammonia/Amines	ppbv	PMS Air Sentry II / Dan Takuma TG-520 SOT-100 series	0.1 ppbv	35x35m (Fab Level) 50x50m (Photo Sub Fab)	1point/3hr
Airborne Molecular Contamination	Condensables* ¹	ppbv	Thermal desorption tubes-Gas Chromatography Mass Spectrometer	0.1 ppbv	PH area	Bi monthly
Airborne Molecular Contamination	Refractories* ²	ppbv	Thermal desorption tubes-Gas Chromatography Mass Spectrometer	0.1 ppbv	PH area	Bi monthly
Airborne Molecular Contamination	Ozone (O3)	ppb	Ecotech Serinus10 SOT-100 series	0.5 ppbv	SOD process area	1point/3hr

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Environment/ Contamination	Parameter	UOM	Analytical Instrumentation	Analytical Detection Limit (DL)	Fab Sampling Location	Sample Frequency & Duration (minimum)
Airborne Molecular Contamination	Hydrogen Sulfide (H ₂ S)	ppb	Ecotech Serinus 51 Horiba ASPA 370-CU1 / Dan Takuma TG-510 / SOT-100 series	3 ppbv	Copper Utility Zone.	1 point/3 hr

*1: Condensables (as defined by ASML) (BP≥150°C, MW≥120)

*2: Refractories (all organic compounds containing any other element than H, C, O, and N)

Appendix III – Recommended Analytical Instrumentation for UPW Characterization

Parameter	Spec Compliant BIC Analytical Instrumentation Type	Best in Class Analytical Detection Limit (DL)	Standard Analytical Instrumentation Type to monitor shifts/trends	Standard DL Capability
Temperature	Thermocouple		Thermocouple	
Resistivity	Resistivity Meter		Resistivity Meter	
Maximum Particle Count	Laser Scatter		Laser Scatter	
Maximum Dissolved Silica	Online IC	100	Online IC	100
Maximum Total Silica	as Si by ICP-MS	5000	Colorimetric or as Si by ICP-MS	5000
Maximum Dissolved Oxygen (DO)	DO Meter		DO Meter	
Maximum Dissolved Nitrogen				
Maximum TOC	TOC Analyzer		TOC Analyzer	
Boron	B-Analyzer; ICP-MS	50	B-Analyzer or ICP-MS	50

Parameter	Spec Compliant BIC Analytical Instrumentation Type	Best in Class Analytical Detection Limit (DL)	Standard Analytical Instrumentation Type to monitor shifts/trends	Standard DL Capability
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H2O2	Titration; IC	0.05	Titration; Thermo ICS-5000/6000 IC	0.05
Bacteria	Culture Counts	N/A	Culture Counts	N/A
Na Cation	IC/ ICP-MS	1	Thermo ICS 5000+ Integral/Agilent 8900 ICP-MS	10
K Cation	IC/ ICP-MS	1	Thermo ICS 5000+ Integral/Agilent 8900 ICP-MS	20
Ammonium	IC	1000	ICS-5000/6000 IC	1000
Ca Cations	IC/ ICP-MS	1	Online IC/ICP-MS	30
Mg Cations	IC/ ICP-MS	1	Online IC/ICP-MS	10
Cu Cations	IC/ ICP-MS	1	Online IC/ICP-MS	5
Fe Cations	IC/ ICP-MS	1	Online IC/ICP-MS	10
Mn Cations	IC/ ICP-MS	1	Online IC/ICP-MS	100
Ni Cations	IC/ ICP-MS	1	Online IC/ICP-MS	5

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Parameter	Spec Compliant BIC Analytical Instrumentation Type	Best in Class Analytical Detection Limit (DL)	Standard Analytical Instrumentation Type to monitor shifts/trends	Standard DL Capability
Zn Cations	IC/ ICP-MS	1	Thermo ICS 5000+ Integral/as Zn ICP-MS	15
Tetra Methyl Ammonium	IC	5	Thermo ICS 5000+ Integral	5
Al Cations	IC/ ICP-MS	1	Thermo ICS 5000+ Integral/as Al ICP-MS	15
Sr	ICP-MS	1	ICP-MS	100
Ba	ICP-MS	1	ICP-MS	100
Cr	ICP-MS	1	ICP-MS	5
Sn	ICP-MS	1	ICP-MS	100
Li	ICP-MS	1	ICP-MS	10
Ag	ICP-MS	1	ICP-MS	20
Pb	ICP-MS	1	ICP-MS	100
Mo	ICP-MS	1	ICP-MS	10
W	ICP-MS	1	ICP-MS	10
Cd	ICP-MS	1	ICP-MS	100
Co	ICP-MS	1	ICP-MS	10
Ti	ICP-MS	1	Agilent 8900 ICP-MS	100
V	ICP-MS	1	ICP-MS	100

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Parameter	Spec Compliant BIC Analytical Instrumentation Type	Best in Class Analytical Detection Limit (DL)	Standard Analytical Instrumentation Type to monitor shifts/trends	Standard DL Capability
Fluoride	IC		Thermo ICS 5000+ Integral	150
Chloride	IC		Thermo ICS 5000+ Integral	500
Bromide	IC		ICS-5000/6000 IC	300
Nitrate	IC		ICS-5000/6000 IC	1000
Nitrite	IC		ICS-5000/6000 IC	100
Phosphate	IC		ICS-5000/6000 IC	1000
Sulfate	IC		Thermo ICS 5000+ Integral	500

POD = Point of Distribution (located downstream of final filtration) and UPW/HUPW Supply to Fab C-Critical,
NC-Non-Critical, E-Engineering/Monitor

Appendix IV - Additional Information on Importance of UPW Parameters Sampling, Frequency and Analysis for Global Specification Validation

SPC type	Parameters	Sampling Method	Sampling Location	Analytical Frequency	Minimum Site Analytical Capability (Instrumentation/Parameter)	Global Spec Validation Frequency	OOS Reaction Mechanism ¹ (Monitor daily for shifts/trends and respond as required)	Notes/Discussion
C	T, resistivity, particle, DO, DN2, TOC, B	online	POD (Point of Distribution)	Continuous for all params	Online analytical must be capable of testing to global spec target ²	Continuous online testing	RM 1	*Exception approval required for interim testing frequency when continuous cannot meet spec; LT plan should be to purchase capable instrument. See Figure 1 workflow
C	Max Dissolved Silica*	online	multiple point in polish & POD	Daily (minimum once/24 hr)	Online or Daily analytical must be capable of testing to global spec target ² *Online IC can be used	Minimum of one sample tested daily	RM 1	*Exception approval required for interim testing frequency when continuous cannot meet spec; LT plan should be to purchase capable instrument. See Figure 1 workflow
C	Na, K, Cu, Fe, Ni, Sn, Co, Zn Max Total Silicon as measured by ICP-MS (UPW and HUPW)	Manual collection /online	All--POD Na, Zn cations- Multiple points in polish and PD	Daily with online IC* to lowest DL available for all params except Si and Sn Quarterly with ICP MS for Sn, and Max Total Si- or on an as needed basis to troubleshoot elevated dissolved Si analysis or particle issues	Online IC to lowest DL available for all Params except Si and Sn Max Total Si: Dedicated In house analytical not required (see notes) All remaining ions: Existing ICP-MS in each of site labs is capable analyzing at global spec detection limit (DL), however background subtraction may be required for Si analysis)	Required quarterly Validate with in house ICP-MS when online IC DL > global spec, or doesn't monitor for analyte (ie. Sn)	RM 1 If system shutdown occurs, what is required for system restart? ICP-MS actuals showing system is operationally meeting global spec for all except total Silicon If shift due to UPW treatment issue, system status- what is the worst operating condition that forces a shutdown? -- contact process to make the call	Max Total Silicon: Analysis used to validate elevated Silica results due to resin breakthrough. Particle size unknown from digestion method. Dissolved Si and B used as indicators A dedicated ICPMS required only when an online method, IC or other, is not available to measure dissolved Silica. Total Silicon is more of an indicator rather than an absolute Ni, Zn, Co- can be analyzed by on-line IC, but on-line IC DL not capable at global spec Sn cannot be tested with on-line IC, requires testing in lab on ICP-MS

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SPC type	Parameters	Sampling Method		Analytical Frequency	Minimum Site Analytical Capability (Instrumentation/Parameter)	Global Spec Validation Frequency	OOS Reaction Mechanism ¹ (Monitor daily for shifts/trends and respond as required)	Notes/Discussion
E	F-, Cl-, PO ₄ ³⁻ , SO ₄ ²⁻ , NO ₂ -, NO ₃ -, Br-, NH ₄ ⁺ TMA (Trimethylamine) TMA+(Tetramethyl ammonium ion)	online/manual collection	TMA (both), F-, Cl-, PO ₄ ³⁻ , SO ₄ ²⁻ multiple point in polish & POD All others POD	Daily (minimum once/24 hr)	Online IC with the lowest DL available <i>(DL is not required to meet global spec since this is an Engineering Parameter)</i>	Quarterly Validate using in house or off site (Balaz/F4 MTI) ICPMS when online IC DL is greater than spec target. **It is critical to follow clean sample bottle prep and sampling technique to prevent sample contamination.**	RM 2 As an E/Monitor param, a system excursion for <i>small shifts/trends would not likely result in a shutdown</i> , but it may be necessary for large swings. This should be determined with notification and review with production process engineering	Recommend: Online IC for each UPW system For successful quarterly grab sample validation use clean sample prep and collection method; When chemistry laboratory analysis exceeds spec refer to critical param results and follow RM2 consulting with production process engineering. <i>TMA+ (tetramethyl ammonium ion) method analysis for Online IC in development</i>
E	Most metals, e.g Cr, W, Ti, Ag,	Manual collection	POD	Quarterly for W only if CMP Reclaim returned to UPW Semi Annual (once/6 months) for all remaining parameters	Analyze using in house ICP-MS capable of DL at global spec limit or send samples quarterly to offsite Lab (e.g. F4) for global spec validation	Tungsten (W) Required quarterly with ICP-MS All Metals required every 6 months	RM 2 Confirm OOS results with additional sampling. Review OOS condition with production process engineering to determine product disposition.	If in house chemistry laboratory ICP-MS not capable of global Spec DL, then results of the Critical SPC parameters (C) would be used to determine system online/restart status, and a sample would be sent offsite spec validation
E	bacteria	Manual collection	POD & Varies	As needed				
E	H2O2	Manual collection	POD	TBD	TBD- method under evaluation			

1) ¹ Reaction Mechanisms

Notes:

- What triggers shutdown when alignment not based on System Design target: Process Engineering responsible to determine what if any system shutdown or specific product disposition required for quality concerns if system unable to return to typical actuals where system meets global specification even when system design specifications do not.
- If shift due to UPW treatment issue, system status- what is the worst operating condition that forces a shutdown? --contact process to make the call.

• **RM1. Critical Parameter Reaction Mechanism**

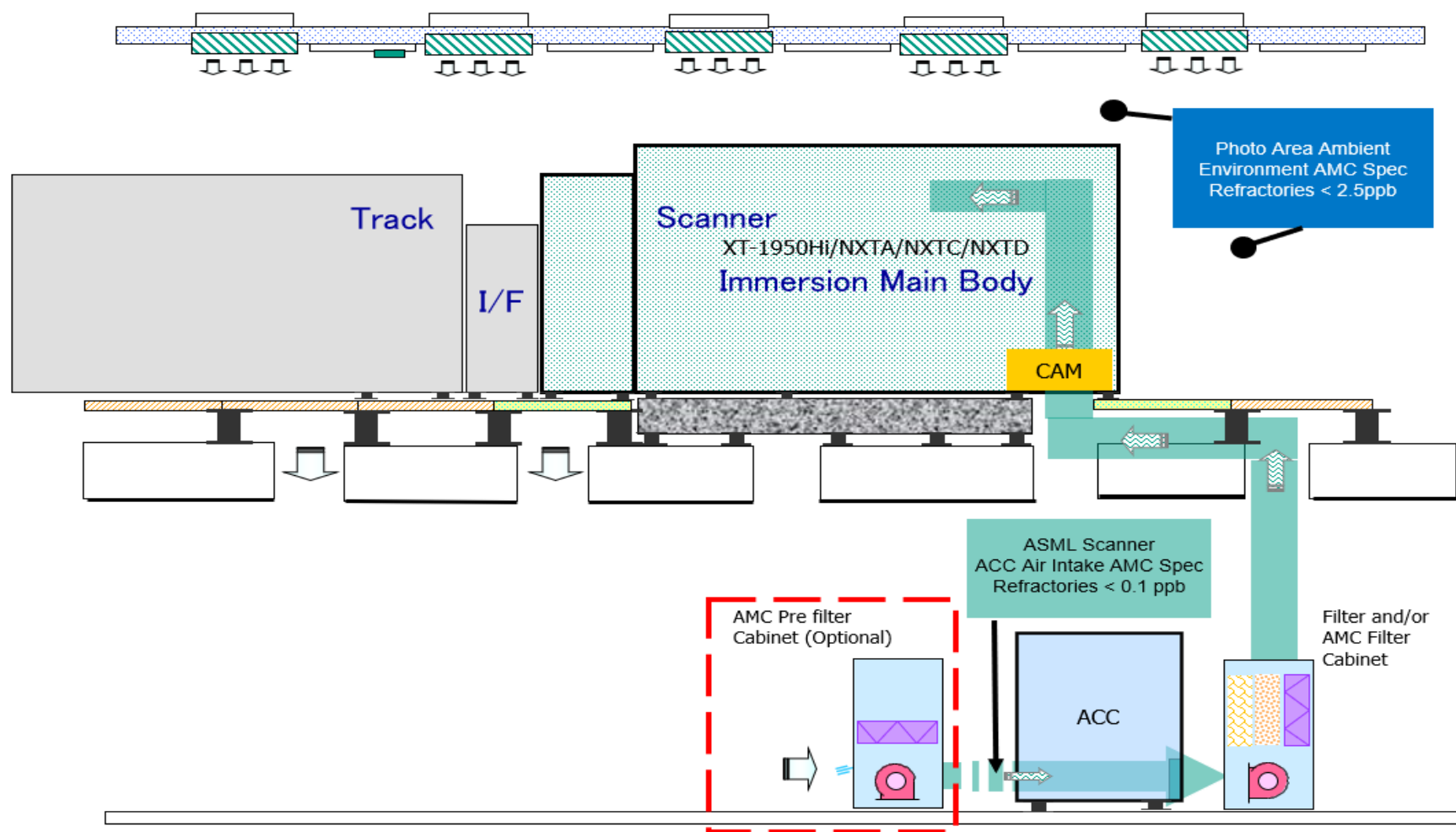
- Validate/correct analytical instrumentation performance
- Check/correct Water Supply System operation
- Notify process if condition persists to determine next steps and when/if a system shutdown is required
- Validate when system returns to specification target with online, or lab sample testing using analytical capable of testing at global spec detection limit

• **RM2. Engineering Param Reaction Mechanism**

- Validate/correct analytical instrumentation performance
- Check/correct Water Supply System operation
- Continue to monitor and notify process if large swings in trends or shift persists to determine next steps and when/if a system shutdown is required
- Validate system returns to specification/typicals with online, or lab sample testing (if onsite lab is not capable of detecting at DL, then use critical params as indicator for system meeting specification, and send sample off site for Spec Validation.

2) ²Capable of testing to meet global specification target- System DL is < global spec target

Appendix V – Required Set-Up for Achieving Critical Specifications for ASML Warranty



Appendix VI – Sample Certificate of Compliance with impurities

**Airborne Labs International, Inc.**220 World's Fair Drive Somerset, NJ 08873 Voice: 732-362-1950 Fax: 732-362-3036
Email: airborne@airlabs.com Website: www.airbornelabs.com

MESSER Gases for Life		Carbon Dioxide (CO₂) Analysis Report	
Location:	Price	Analysis No:	20-0485
Address:	1179 East Main Street/PMB 102	Sample Date:	06/03/20
City, State, Zip Code:	Price, UT 84501	Sample Received:	06/04/20
Attention:	Travis Allen	Final Test Date:	06/09/20
Telephone:	435-613-0613	Report Date:	06/10/20
FAX:	435-613-0612		
e-mail:	travis.allen@messer-us.com; Paul.burke@messer-us.com		
cc:	Joe.Angeloni@messer-us.com;		
Analysis:	Product	Analysis Type:	Quarter 2
Sample ID:	5674-B-060320-1		
Sample ID:	Received in a Messer No-Haz Kit.		
[REDACTED]		[REDACTED]	
[REDACTED]		[REDACTED]	
[REDACTED]		[REDACTED]	
[REDACTED]		[REDACTED]	
[REDACTED]		[REDACTED]	
Test Description (Units)			
CO ₂ Purity (% v/v):	99.99+	DL	90
CO ₂ Purity Method:	by Difference		
Water Vapor, Moisture (ppm v/v):	1.8		1
Hydrogen (ppm v/v):	nd		1
Oxygen + Argon (ppm v/v):	1.9		1
Oxygen (ppm v/v):	--		1
Argon (ppm v/v):	--		--
Nitrogen (ppm v/v):	4.0		1
Carbon Monoxide (ppm v/v):	nd		1
Ammonia, NH ₃ (ppm v/v):	nd		0.5
Oxides of Nitrogen (ppm v/v):	nd		0.5
Nitric Oxide, NO (ppm v/v):	--		0.5
Nitrogen Dioxide, NO ₂ (ppm v/v):	--		0.5
NVR, Total Non-Volatile Residue (ppm w/w):	nd		2
NVOR, Total Non-Volatile Organic Residue (ppm w/w):	nd		2
Phosphine (PH ₃) (ppm v/v):	--		0.25
Total Hydrocarbons, THC (ppm v/v as CH ₄):	0.2		0.1
Total Non-Methane Hydrocarbons, THC - CH ₄ (ppm v/v as CH ₄):	--		--
Acetaldehyde, AA (ppm v/v):	nd		0.05
Total Volatile Sulfur Compounds, TVSC (ppm v/v):	nd		--
Total Sulfurs, TVSC - SO ₂ (ppm v/v):	nd		0.01
Total Other Volatile Oxygenates TVO - AA - MeOH - EtOH (ppm v/v):	--		--
Aromatic Hydrocarbons (BTEX, ppb v/v)			
Benzene or Aromatic HC Content as Benzene:	nd		2
Toluene:	--		2
Ethyl Benzene:	--		2
o+m+p Xylenes:	--		2



Airborne Labs International, Inc.

220 World's Fair Drive - Somerset, NJ 08873 - Voice: 732-382-1950 Fax: 732-382-3036
Email: airborne@alabs.com Website: www.airbornelabs.com



Carbon Dioxide (CO₂) Analysis Report

Supplemental Tests (ppm v/v)	Results	DL
Hydrogen Cyanide, HCN:	nd	0.2
Vinyl Chloride:	nd	0.1
Volatile Hydrocarbons (VHC, ppm v/v)		
Methane:	nd	0.1
Ethylene:	--	0.1
Ethane:	--	0.1
Propylene:	--	0.1
Propane:	--	0.1
Isobutane:	--	0.1
n-Butane:	--	0.1
Butenes:	--	0.1
Isopentane:	--	0.1
n-Pentane:	--	0.1
Pentenes:	--	0.1
Hexanes+:	--	0.1
Volatile Sulfur Compounds (VSC, ppm v/v)		
Hydrogen Sulfide (H ₂ S):	--	0.01
Carbonyl Sulfide (COS):	--	0.01
Sulfur Dioxide (SO ₂):	nd	0.01
Methyl Mercaptan:	--	0.01
Ethyl Mercaptan:	--	0.01
Dimethyl Sulfide:	--	0.01
Carbon Disulfide:	--	0.01
i-Propyl Mercaptan:	--	0.01
t-Butyl Mercaptan:	--	0.01
n-Propyl Mercaptan:	--	0.01
Methyl Ethyl Sulfide:	--	0.01
sec-Butyl Mercaptan:	--	0.01
i-Butyl Mercaptan:	--	0.01
Diethyl Sulfide:	--	0.01
n-Butyl Mercaptan:	--	0.01
Dimethyl Disulfide:	--	0.01
Diethyl Disulfide:	--	0.01
Other Volatile Sulfur Compounds:	--	0.01



Airborne Labs International, Inc.

220 World's Fair Drive Somerset, NJ 08873 Voice: 732-362-1950 Fax: 732-362-3008
Email: airborne@airlabs.com Website: www.airbornelabs.com

MESSER Gases for Life		
Carbon Dioxide (CO ₂) Analysis Report		
Volatile Oxygenates (VOX, ppm v/v)	Results	DL
Ethylene Oxide:	--	0.1
Dimethyl Ether:	--	0.1
Diethyl Ether:	--	0.1
Methyl Ethyl Ether:	--	0.1
Methanol:	nd	0.1
Propionaldehyde:	--	0.1
Acetone:	--	0.1
Ethanol:	--	0.1
Isopropanol:	--	0.1
Ethyl Acetate:	--	0.1
Methyl Ethyl Ketone:	--	0.1
t-Butanol:	--	0.1
n-Propanol:	--	0.1
2-Butanol:	--	0.1
Isobutanol:	--	0.1
n-Butanol:	--	0.1
Isoamyl Alcohol:	--	0.1
Isoamyl Acetate:	--	0.1
Unknown Volatile Oxygenates:	--	0.1
Comments:	This sample meets all Messer purity requirements for LCO ₂ .	

Reviewed By: Jeff Wahome 06/10/20
Authorized Signature / Date

Analytical Operations Manager
Title

Airborne Labs International

NOTES: DL = test detection limit; nd = less than test detection limit; na = test not performed; sc = see comments; % = parts per hundred; ppm = parts per million; ppb = parts per billion; v/v = volume analyte/volume sample; w/w = weight analyte/weight sample; G.T., + = greater than amount specified; Unit Conversions: 1 ppm v/v = 1 uL/L = 1000 ppb v/v = 0.0001% v/v.